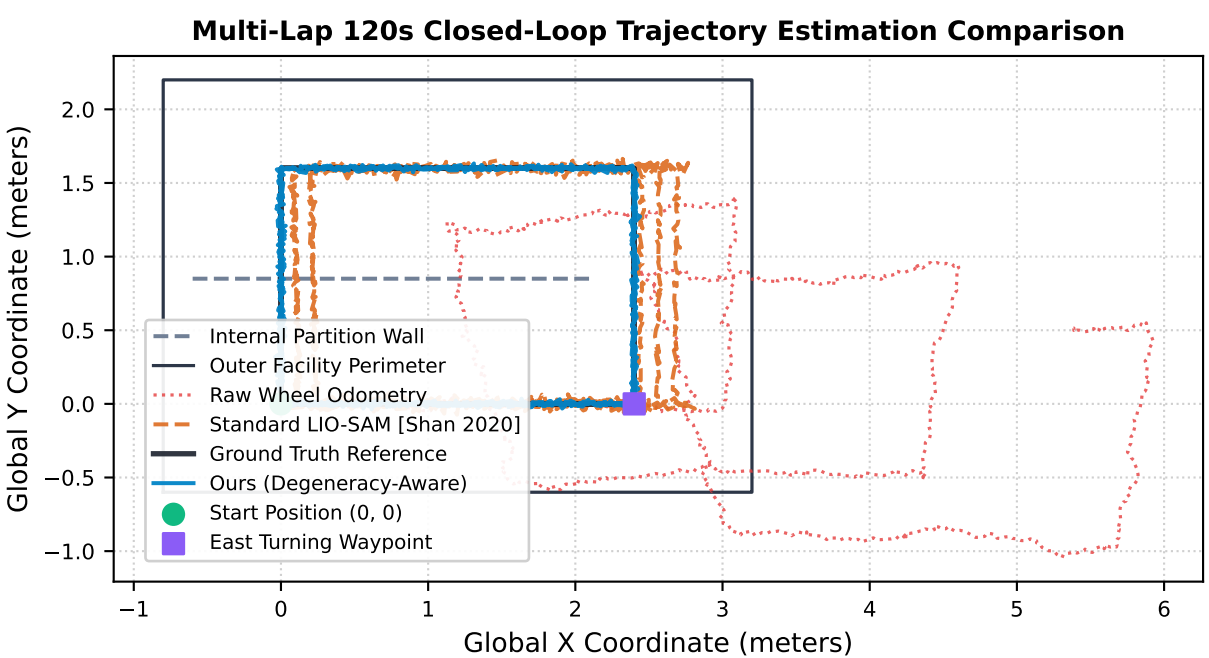

Multi-Lap 120s Closed-Loop Trajectory Estimation Comparison
Global X Coordinate (meters)
Global Y Coordinate (meters)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1
0
1
2
3
4
5
6
Internal Partition Wall
Outer Facility Perimeter
Raw Wheel Odometry
Standard LIO-SAM [Shan 2020]
Ground Truth Reference
Ours (Degeneracy-Aware)
Start Position (0, 0)
East Turning Waypoint